

5G RAN

Co-MM (Low-Frequency TDD) Feature Parameter Description

Issue	Draft B
Date	2026-01-30



Copyright © Huawei Technologies Co., Ltd. 2026. All rights reserved.

No part of this document may be reproduced or transmitted in any form or by any means without prior written consent of Huawei Technologies Co., Ltd.

Trademarks and Permissions



HUAWEI and other Huawei trademarks are trademarks of Huawei Technologies Co., Ltd.

All other trademarks and trade names mentioned in this document are the property of their respective holders.

Notice

The purchased products, services and features are stipulated by the contract made between Huawei and the customer. All or part of the products, services and features described in this document may not be within the purchase scope or the usage scope. Unless otherwise specified in the contract, all statements, information, and recommendations in this document are provided "AS IS" without warranties, guarantees or representations of any kind, either express or implied.

The information in this document is subject to change without notice. Every effort has been made in the preparation of this document to ensure accuracy of the contents, but all statements, information, and recommendations in this document do not constitute a warranty of any kind, express or implied.

Huawei Technologies Co., Ltd.

Address: Huawei Industrial Base
Bantian, Longgang
Shenzhen 518129
People's Republic of China

Website: <https://www.huawei.com>

Email: support@huawei.com

Security Declaration

Vulnerability

Huawei's regulations on product vulnerability management are subject to the *Vul. Response Process*. For details about this process, visit the following web page:

<https://www.huawei.com/en/psirt/vul-response-process>

For vulnerability information, enterprise customers can visit the following web page:

<https://securitybulletin.huawei.com/enterprise/en/security-advisory>

Contents

1 Change History.....1

2 About This Document.....4

2.1 General Statements.....4

2.2 Features in This Document.....4

2.3 Differences.....5

3 Overview.....6

4 Co-MM.....7

4.1 Principles.....7

4.1.1 Over-the-Air Channel Calibration.....8

4.1.2 Coherent Joint Transmission.....9

4.2 Network Analysis.....15

4.2.1 Benefits.....15

4.2.2 Impacts.....16

4.3 Requirements.....17

4.3.1 Licenses.....17

4.3.2 Functions.....17

4.3.2.1 Prerequisite Functions.....18

4.3.2.2 Mutually Exclusive Functions.....19

4.3.2.3 Function Impacts.....23

4.3.3 Hardware.....38

4.3.4 Cells.....39

4.3.5 Others.....39

4.4 Operation and Maintenance.....39

4.4.1 Data Configuration.....39

4.4.1.1 Data Preparation.....40

4.4.1.2 Using MML Commands.....44

4.4.1.3 Using the MAE-Deployment.....46

4.4.2 Activation Verification.....47

4.4.3 Network Monitoring.....47

5 Parameters.....49

6 Counters.....51

7 Glossary..... 52

8 Reference Documents.....53

1 Change History

The following describes the specific changes in different issues.

5G RAN10.1 Draft B (2026-01-30)

This issue includes the following changes.

- Technical changes
None
- Editorial changes
Revised other descriptions in the document.

5G RAN10.1 Draft A (2025-12-31)

This issue introduces the following changes to 5G RAN9.1 05 (2025-12-14).

- Technical changes

Change Description	Parameter Change	RAT	Base Station Model
Added the impact relationship between Co-MM and time-domain power saving BWP (low-frequency TDD). For details, see 4.3.2.3 Function Impacts .	None	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Added the impact relationship between L/NR Inter-band Deep Coordination and Co-MM. For details, see 4.3.2.3 Function Impacts .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)

Change Description	Parameter Change	RAT	Base Station Model
Added the impact relationship between self-fusion weights and enhanced Co-MM. For details, see 4.3.2.3 Function Impacts .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)
Added the impact relationship between RF module fast deep dormancy and Co-MM. For details, see 4.3.2.3 Function Impacts .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)
Added the impact relationship between uplink 4-layer transmission for a single FWA user and Co-MM. For details, see 4.3.2.3 Function Impacts .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)
Added support for optimized cooperating cell selection. For details, see: • 4.1.2 Coherent Joint Transmission • 4.3.2.1 Prerequisite Functions • 4.4.1.1 Data Preparation • 4.4.1.2 Using MML Commands	Modified parameter: Added the COOP_CELL_ALLO C_OPT_SW option to the NRDUCellComp.C ompOptSwitch parameter.	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Updated the mutually exclusive relationships of Co-MM with Extended Cell Range (TDD) and Extended Cell Range of 90 km (TDD). For details, see 4.3.2.2 Mutually Exclusive Functions .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)

- Editorial changes

Added the configuration requirements when the AAU adopts inter-BBU SDR networking. For details, see [4.3.5 Others](#).

Added descriptions of RF modules required for Co-MM. For details, see [4.3.3 Hardware](#).

Revised other descriptions in the document.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Features in This Document

This document describes the following features.

Feature ID	Feature Name	Chapter/Section
FOFD-070203	Coordinated Massive MIMO (Co-MM)	4 Co-MM

2.3 Differences

Differences Between NSA and SA

Function Name	Difference	Chapter/Section
Co-MM	None	4 Co-MM

Differences Between High Frequency Bands and Low Frequency Bands

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

Function Name	Difference	Chapter/Section
Co-MM	Supported only in low frequency bands	4 Co-MM

3 Overview

Intra-frequency networking is typically used for NR to improve spectrum resource utilization. On such a network, wanted signals for cell-edge UEs are weak and inter-cell interference is high. As a result, user experience deteriorates significantly. To address the user experience deterioration for cell-edge UEs, a coordination technology can be used to allow for coordinated transmission from multiple neighboring transmission reception points (TRPs) to a single UE.

In effect, multi-TRP coordinated transmission involves a larger number of transmit antennas than single-TRP transmission, offering the diversity or spatial multiplexing gains brought by multi-antenna transmission.

To further obtain an increase in antenna array gains, joint channel calibration of the TRPs is required. It achieves downlink-uplink reciprocity between inter-TRP transmit and receive channels and therefore allows for joint channel measurement and beamforming, which improve the performance of joint transmission. **Such coordinated PDSCH data transmission to a single UE based on joint beamforming for TRPs is called Coordinated Massive MIMO (Co-MM).**

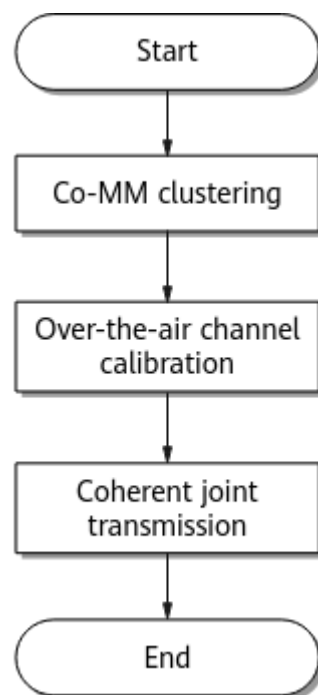
4 Co-MM

4.1 Principles

Co-MM allows for joint transmission from two TRPs of a gNodeB to a single UE. In essence, the two N -antenna TRPs are equivalent to one $2N$ -antenna TRP. This type of transmission increases antenna array gains.

This function is controlled by the **INTRA_GNB_SU_COMM_SW** option of the **NRDUCellCollabServ.MultiCellMimoSwitch** parameter. [Figure 4-1](#) shows how it works.

Figure 4-1 Co-MM workflow



1. Co-MM clustering

A Co-MM cluster is a set of intra-frequency cells that are served by the same base station. Co-MM performs over-the-air channel calibration in individual Co-MM clusters.

Usually, TRPs with coverage overlaps form a cluster and over-the-air channel calibration is performed between these TRPs. Only if the calibration is successful between two TRPs can coherent joint transmission take effect for them.

Co-MM clustering is required before Co-MM is enabled. There are two clustering modes:

- MML-based: A cluster is added by running the MML command **ADD GNBCLUSTER**, and an intra-frequency intra-base-station cell is added to a cluster by running the MML command **ADD GNBMIMOCLUSTERCELL**.
- MAE-based: The NR intelligent cell clustering function on the MAE-Optimization is used. For details, see **NR Intelligent Cell Clustering** under **Description > Function Description > MAE-Optimization Function Description** of *iMaster MAE Product Documentation (EulerOS, TaiShan)*.

2. Over-the-air channel calibration

Joint beamforming for the TRPs requires inter-TRP over-the-air channel calibration, in which the Tx/Rx ratio is adjusted to be identical for two TRPs, to achieve downlink-uplink reciprocity between inter-TRP transmit and receive channels.

3. Coherent joint transmission

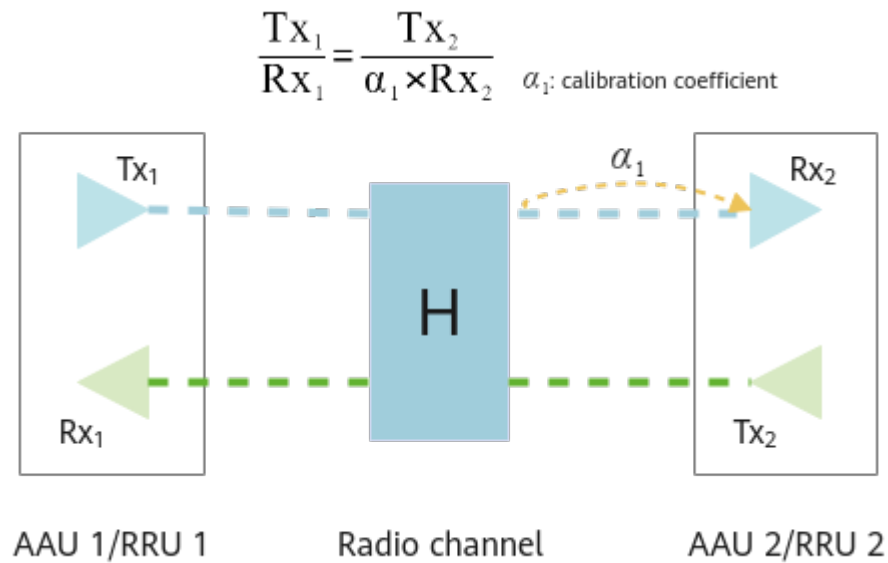
Coherent joint transmission from two TRPs to a UE is a type of data transmission with joint beamforming for the TRPs. It yields higher antenna array, power, and spatial multiplexing gains and increases the downlink user-perceived rates of UEs in overlapping areas.

Over-the-air channel calibration and coherent joint transmission are key technologies of Co-MM, which will be detailed in the subsequent sections.

4.1.1 Over-the-Air Channel Calibration

Figure 4-2 illustrates how over-the-air channel calibration works. Basically, it determines the difference in delay between the channels of different AAUs/RRUs based on the responses of these channels to calibration signals, and then it performs compensation to achieve downlink-uplink reciprocity between inter-TRP transmit and receive channels.

Figure 4-2 Over-the-air channel calibration

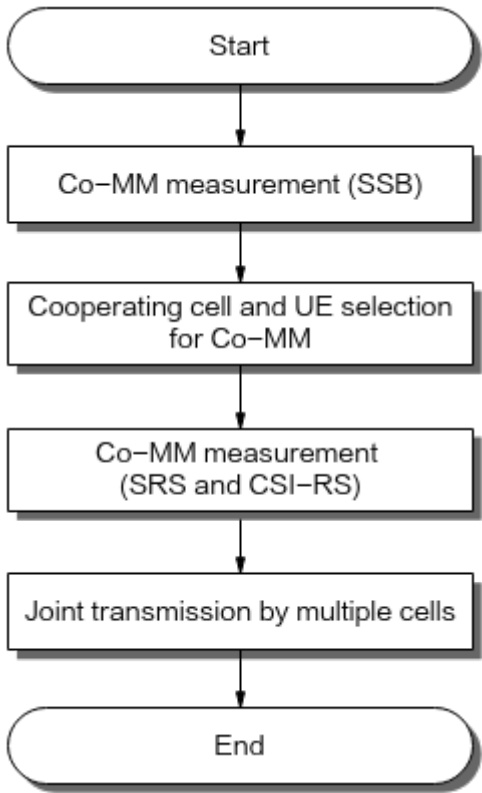


For details about how to check whether inter-AAU/RRU over-the-air channel calibration is successful, see [4.4.2 Activation Verification](#).

4.1.2 Coherent Joint Transmission

Coherent joint transmission from two TRPs to a UE is a type of data transmission with joint beamforming for the TRPs. It yields higher antenna array, power, and spatial multiplexing gains and increases the downlink user-perceived rates of UEs in overlapping areas. [Figure 4-3](#) shows its procedure.

Figure 4-3 Procedure for coherent joint transmission



1. Co-MM measurement (SSB)

After setting up a service, a UE performs synchronization signal and PBCH block (SSB) RSRP measurements. It sends A3 measurement reports when the condition for event A3 is met.

Event A3 indicates that the signal quality of a neighboring cell is higher than that of the serving cell by a certain offset throughout a period of time. A UE reports event A3 if its signal quality meets the condition $(Mn + Ofn + Ocn - Hys > Ms + Ofs + Ocs + Off)$ throughout *TimeToTrigger*. [Table 4-1](#) explains the variables and lists the parameters used to control them.

Table 4-1 Meanings of variables in event A3 and parameters used to control them

Variable	Meaning	Parameter
<i>TimeToTrigger</i>	Period throughout which the entering condition must be met before the event is reported	NRCellComp. <i>TimeToTrigger</i>
<i>Ms</i> and <i>Mn</i>	Measurement results of the serving and neighboring cells, respectively	None
<i>Ofs</i> and <i>Ofn</i>	Frequency-specific offsets of the serving and neighboring cells, respectively. The frequency-specific offset is 0 for intra-frequency cells.	None

Variable	Meaning	Parameter
<i>Ocs</i> and <i>Ocn</i>	Cell individual offsets (CIOs) of the serving and neighboring cells, respectively	The value of <i>Ocs</i> is 0, and <i>Ocn</i> is specified by the NRCellRelation.CellIndividualOffset parameter.
<i>Hys</i>	Hysteresis for the event	NRCellComp.Hysteresis
<i>Off</i>	Offset for the event	NRCellComp.DlCompRsrpOffset

For more details about event A3, see section 5.5 "Measurements" in 3GPP TS 38.331 V17.4.0.

2. Cooperating cell and UE selection for Co-MM

Co-MM requires that joint data transmission to a UE from the serving and cooperating cells use the same time-frequency resources in these cells. To this end, the cooperating cell must be lightly loaded and meet certain TX/RX mode and parameter setting requirements. The detailed requirements for the cooperating cell are as follows:

- Co-MM is enabled in the cell.
- The PRB usage of the cell is less than the value of **NRDUCellComp.DlCompCoCellPrbThld**.
- The number of available downlink RBs in the cell is the same as that in the serving cell.
- The TX/RX mode of the cell meets the requirements described in [Table 4-2](#).

Table 4-2 Requirements for the TX/RX modes of the serving and cooperating cells

TX/RX Mode of the Serving Cell	TX/RX Mode of the Cooperating Cell	Description
32T32R or 64T64R	32T32R or 64T64R	This is an outdoor macro-macro networking scenario.
4T4R	4T4R, 32T32R, or 64T64R	The combinations of 4T4R and massive MIMO work only in outdoor pole site + macro base station scenarios.
8T8R	8T8R, 32T32R, or 64T64R	The combinations of 8T8R and massive MIMO work only in outdoor pole site + macro base station scenarios.

- The parameters listed in [Table 4-3](#) of the cell have the same settings as those of the serving cell.

Table 4-3 Parameters required to be with the same settings between the serving and cooperating cells

Parameter Name	Parameter ID	Option
NR DU Cell Networking Mode	NRDUCell.NrDuCellNetworkingMode	None
Frequency Band	NRDUCell.FrequencyBand	None
Downlink NARFCN	NRDUCell.DlNarfcn	None
Downlink Bandwidth	NRDUCell.DlBandwidth	None
Slot Assignment	NRDUCell.SlotAssignment	None
Cyclic Prefix Length	NRDUCell.CyclicPrefixLength	None
Downlink DMRS Configuration Type	NRDUCellPdsch.DlDmrsConfigType	None
Downlink DMRS Maximum Length	NRDUCellPdsch.DlDmrsMaxLength	None
Downlink Additional DMRS Position	NRDUCellPdsch.DlAdditionalDmrsPos	None
Common Control Resource RB Number	NRDUCellCoreset.CommonCtrlResRbNum	None
TRS Period	NRDUCellCsirs.TrsPeriod	None
Coordination TRS Resource Switch	NRDUCellAlgoSwitch.CoordTrsResourceSwitch	JT_TRS_RES_SW
Max CSI-RS Power Offset	NRDUCellChnPwr.CsirsMaxPwrOffset	None
DL CoMP Algorithm Switch	NRDUCellComp.DlCompAlgoSwitch	CSIRS_SCH_ENH_SW
Multi-Cell MIMO Switch ^a	NRDUCellCollabServ.MultiCellMimoSwitch	INTER_CLUSTER_SU_COMM_SW
a: The settings of the INTER_CLUSTER_SU_COMM_SW option of the NRDUCellCollabServ.MultiCellMimoSwitch parameter must be identical between the serving and cooperating cells only when the CSIRS_SCH_ENH_SW option of the NRDUCellComp.DlCompAlgoSwitch parameter is deselected.		

After receiving A3 reports, the gNodeB selects the intra-base-station neighboring cell with the best signal quality from the reported cells that meet all the preceding requirements as a cooperating cell for Co-MM. The UE that sends the A3 reports is regarded as the Co-MM UE. If no neighboring cell meets the preceding requirements, the procedure ends.

In scenarios with dense deployment of multiple cells, Co-MM may fail to take effect if no suitable cooperating cell can be selected. To address this issue, you are advised to select the **COOP_CELL_ALLOC_OPT_SW** option of the **NRDUCellComp.CompOptSwitch** parameter to expand the range of cooperating cells that can be selected in the overlapping area, thereby increasing the probability of Co-MM taking effect.

3. Co-MM measurement (SRS and CSI-RS)

The gNodeB performs SRS measurements in the serving and cooperating cells of the identified Co-MM UE. If the cooperating cell cannot offer the SRS resources corresponding to those allocated in the serving cell of the UE, Co-MM cannot take effect.

After identifying the Co-MM UE, the gNodeB instructs the UE to perform CSI-RS measurement.

4. Joint transmission by multiple cells

When the serving cell determines that the SRS SINR in the cooperating cell for the Co-MM UE is greater than threshold 1 and one of the following conditions is met, the serving cell instructs the cooperating cell to perform coherent joint transmission for the UE, improving the receive signal quality for the UE:

- SRS RSRP in the cooperating cell – SRS RSRP in the serving cell > Threshold 2
- SRS RSRP in the cooperating cell > Threshold 3

In addition, the spectral efficiency is continuously compared between different measurement modes. If the ratio of the spectral efficiency obtained based on joint measurement to that obtained based on single-cell measurement is less than threshold 4, coherent joint transmission switches to single-cell transmission. In single-cell transmission, if the ratio of the spectral efficiency obtained based on single-cell measurement to that obtained based on joint measurement is less than threshold 4, single-cell transmission quickly switches back to coherent joint transmission.

Table 4-4 lists the parameters that specify the preceding thresholds.

Table 4-4 Parameters that specify the thresholds

Threshold	Parameter
Threshold 1	NRDUCellCollabServ.CoopCellPreSnrThld
Threshold 2	NRDUCellCollabServ.SrsRsrpDiffThld
Threshold 3	NRDUCellCollabServ.CoCellRsrpThld
Threshold 4	NRDUCellComp.JtQuickRollbackSpectEffThld

The functions described in **Table 4-5** can be used to improve the performance of coherent joint transmission.

Table 4-5 Joint transmission performance improvements

Function	Description	Parameter
Joint transmission delay compensation	Delay compensation is performed if the absolute value of the timing advance (TA) measurement difference between the serving and cooperating cells is greater than or equal to the value of the relevant parameter.	NRDUCellComp.JtDelayCompensationThld
Joint transmission evaluation optimization	The joint transmission evaluation algorithm is optimized to improve the accuracy of joint transmission evaluation results.	JT_JUDGE_OPT_SW option of the NRDUCellComp.DlCompAlgoSwitch parameter
SRS-based weight enhancement for joint transmission	The gNodeB performs SRS-based weight enhancement for joint transmission to reduce inter-stream interference between the serving and cooperating cells.	JT_SRS_WEIGHT_ENH_SW option of the NRDUCellComp.DlCompAlgoSwitch parameter
Weight optimization for Co-MM	Weight optimization is performed for Co-MM UEs in the case of abnormal cross-correlation.	WEIGHT_OPT_SW option of the NRDUCellCollabServ.CommAlgoSwitch parameter
Phase compensation for Co-MM	This function compensates for the phase difference caused by the frequency deviation between the serving and cooperating cells of a Co-MM UE in mobility, increasing the average downlink throughput of the UE.	COMM_PHASE_COMPENSATION_SW option of the NRDUCellCollabServ.CommAlgoSwitch parameter
Rank adaptation for joint transmission	Based on the UE-reported rank indication (RI), the gNodeB selects a rank that better matches the signal quality for data transmission.	JT_RANK_ADAPT_SW option of the NRDUCellComp.DlCompAlgoSwitch parameter

Function	Description	Parameter
Enhanced Co-MM	It increases the proportion of UEs for which Co-MM takes effect in overlapping areas and further lifts the downlink throughput of UEs in the overlapping areas. As this function requires that UEs be allocated SRS resources, it is recommended that SRS interference coordination based on self-contained slots (controlled by the SRS_INTRF_COORD_S_SLOT_SW option of the NRDUCellSrs.SrsAlgoExtSwitch parameter) be enabled.	INTRA_GNB_SU_CO_MM_ENH_SW option of the NRDUCellCollabServ.MultiCellMimoSwitch parameter
Delayed first-packet scheduling	This function delays first-packet scheduling for UEs, making Co-MM more likely to take effect.	FIRST_PACKET_SCH_DELAY_SW option of the NRDUCellCollabServ.CommAlgoSwitch parameter

4.2 Network Analysis

4.2.1 Benefits

On an intra-frequency network with continuous coverage, Co-MM increases the receive power and the number of receive paths for UEs in overlapping areas between intra-base-station cells when the network load is light. This lifts the downlink throughput of these UEs. The fewer the receive paths for UEs in overlapping areas between intra-base-station cells before Co-MM takes effect, the more significant the gains. The larger the traffic volume of these UEs, the more significant the gains.

When common cells and cells enabled with Fusion Cell are mixed and Co-MM is enabled for the common cells, the function of interference avoidance for inter-TRP calibration in a cell enabled with Fusion Cell (controlled by the **FUSION_CALIB_INTRF_AVOID_SW** option of the **gNodeBAlgo.ChnCalibPolSw** parameter) can be used to avoid calibration interference between Fusion Cell clusters and Co-MM clusters. This function increases the success rate of inter-TRP over-the-air channel calibration and therefore lifts the average downlink throughput of UEs in overlapping areas. For details about this function and cells enabled with Fusion Cell, see *Fusion Cell (Low-Frequency TDD)*.

To improve the gains of Co-MM, downlink precise RB estimation (controlled by the **PRECISE_DL_RB_ESTIMATE_SW** option of the **NRDUCellDISch.DISchAlgoSwitch** parameter) is recommended. For details about this function, see *Downlink Scheduling*.

For details about how to evaluate the gains brought by this function, see [4.4.3 Network Monitoring](#).

4.2.2 Impacts

The impacts between this function and other functions are described in [4.3.2.3 Function Impacts](#).

Co-MM increases the downlink spectral efficiency of UEs in the intra-base-station overlapping areas. The average downlink rank and average downlink MCS index may change, and the downlink initial block error rate (IBLER) and residual block error rate (RBLER) may rise.

With Co-MM enabled, idle resources in the local cell can be used for downlink joint transmission. As a result, the downlink PRB usage may increase.

When the **INTRA_GNB_SU_COMM_ENH_SW** option of the **NRDUCellCollabServ.MultiCellMimoSwitch** parameter is selected, the counter (average number of used downlink PRBs) counts in the counter (average number of PRBs used for Co-MM in the local cell offering collaboration for its neighboring cells). The difference between the values of the two counters is the total number of downlink PRBs used for all UEs that have accessed the local cell.

Delayed first-packet scheduling increases the first-packet scheduling delay.

For the definitions of the related indicators, see [Table 4-6](#).

Table 4-6 Definitions of indicators involved in network impacts

Indicator	Definition
Average downlink rank	$\frac{\text{SUM}(k \times \text{N.PDSCH.InitTbDL.Rank}.k)}{\text{SUM}(\text{N.PDSCH.InitTbDL.Rank}.k)}$ $k = 1 \text{ to } 4$
Average downlink MCS index	$\frac{\text{SUM}(k \times \text{N.ChMeas.PDSCH.MCS}.k)}{\text{SUM}(\text{N.ChMeas.PDSCH.MCS}.k)}$ $k = 0 \text{ to } 28$
Downlink IBLER	$\frac{(\text{N.DL.SCH.QPSK.ErrTB.Ibler} + \text{N.DL.SCH.16QAM.ErrTB.Ibler} + \text{N.DL.SCH.64QAM.ErrTB.Ibler} + \text{N.DL.SCH.256QAM.ErrTB.Ibler})}{(\text{N.DL.SCH.QPSK.TB} + \text{N.DL.SCH.16QAM.TB} + \text{N.DL.SCH.64QAM.TB} + \text{N.DL.SCH.256QAM.TB})} \times 100\%$
Downlink residual retransmission BLER	$\frac{(\text{N.DL.SCH.QPSK.ErrTB.Rbler} + \text{N.DL.SCH.16QAM.ErrTB.Rbler} + \text{N.DL.SCH.64QAM.ErrTB.Rbler} + \text{N.DL.SCH.256QAM.ErrTB.Rbler})}{(\text{N.DL.SCH.QPSK.TB} + \text{N.DL.SCH.16QAM.TB} + \text{N.DL.SCH.64QAM.TB} + \text{N.DL.SCH.256QAM.TB})} \times 100\%$

Indicator	Definition
Downlink PRB usage	$N.PR.B.DL.Used.Avg / N.PR.B.DL.Avail.Avg \times 100\%$
Average number of used downlink PRBs in the serving cell	N.PR.B.DL.Used.Avg
Average number of PRBs used for coordinated transmission for Co-MM in the local cell for neighboring cells	N.PR.B.DL.CellCoordinatedUsed.Avg
First-packet scheduling delay	N.Traffic.DL.RlcFirstPktDelay.Time

4.3 Requirements

4.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-070203	Coordinated Massive MIMO (Co-MM)	NR0S00COMM00	Per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.2.1 Prerequisite Functions

Function Name	Function Switch	Reference	Description
TRS rate matching	TRS_RATEMATCH_SW option of the NRDUCellPdsch.RateMatchSwitch parameter	<i>Downlink Scheduling</i>	Enabling Co-MM affects the use of TRS resources. To prevent this impact, the gNodeB reserves TRS resources for cells after TRS rate matching is enabled. TRS rate matching or narrowband TRS rate matching must be enabled before Co-MM is enabled.
Narrowband TRS rate matching	NARROW_BAND_TRS_RATEMATCH_SW option of the NRDUCellPdsch.RateMatchSwitch parameter	None	
Slot configuration	NRDUCell.SlotAssignment	<i>Standards Compliance</i>	The NRDUCell.SlotAssignment parameter must be set to 4_1_DDDSU , 8_2_DDDDDDDSUU , or 8_2_DDDSUUUUUU . The number of GP symbols in a self-contained slot must be within the range of 2 to 6. For details about the setting of the number of GP symbols in self-contained slots with different slot configurations, see <i>Standards Compliance</i> .
Co-MM	INTRA_GNB_SU_COMM_SW option of the NRDUCellCollabServ.MultiCellMimoSwitch parameter	<i>Co-MM (Low-Frequency TDD)</i>	Co-MM and UE-characteristic-based SRS period adaptation must be enabled before enhanced Co-MM (controlled by the INTRA_GNB_SU_COMM_ENH_SW option of the NRDUCellCollabServ.MultiCellMimoSwitch parameter) is enabled.
UE-characteristic-based SRS period adaptation	USER_CHARACTER_SRS_ADAPT_SW option of the NRDUCellSrs.SrsAlgorithm parameter	<i>Channel Management</i>	

Function Name	Function Switch	Reference	Description
Uplink smart preallocation	UL_SMART_PREALLOCATION_SWITCH option of the NRDUCellPusch.UlPreallocationSwitch parameter	<i>Uplink Scheduling</i>	Phase compensation for Co-MM (controlled by the COMM_PHASE_COMPENSATION_SW option of the NRDUCellCollabServ.CommAlgoSwitch parameter) can be enabled only when uplink smart preallocation is enabled.
Enhanced Co-MM	INTRA_GNB_SU_COMM_ENH_SW option of the NRDUCellCollabServ.MultiCellMimoSwitch parameter	<i>Co-MM (Low-Frequency TDD)</i>	Delayed first-packet scheduling (controlled by the FIRST_PACKET_SCH_DELAY_SW option of the NRDUCellCollabServ.CommAlgoSwitch parameter) can be enabled only if enhanced Co-MM is enabled.
SRS interference coordination based on self-contained slots	SRS_INTRF_COORD_SELF_SLOT_SW option of the NRDUCellSrs.SrsAlgoExtSwitch parameter	None	Optimized cooperating cell selection (controlled by the COOP_CELL_ALLOC_OPT_SW option of the NRDUCellComp.CompOptSwitch parameter) requires that SRS interference coordination based on self-contained slots be enabled first.

4.3.2.2 Mutually Exclusive Functions

Function Name	Function Switch	Reference	Description
Hyper Cell	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL	<i>Hyper Cell</i>	Co-MM is not supported in hyper cell networking.

Function Name	Function Switch	Reference	Description
Cell Combination and Joint Transmission Cell Combination	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL_COMBINE_MODE	<i>Cell Combination</i>	Co-MM is not supported in Cell Combination or Joint Transmission Cell Combination networking.
High-speed Railway Superior Experience	NRDUCell.HighSpeedFlag set to HIGH_SPEED	<i>High Speed Mobility</i>	Co-MM is not supported in high-speed scenarios.
DL Blanking	NRDUCell.PeerCellSlotAssignment	<i>Standards Compliance</i>	When Co-MM is enabled, the NRDUCell.PeerCellSlotAssignment parameter can only be set to NOT_CONFIG .
U-TDOA-based positioning/ TOA fingerprint-based positioning	UTDOA_SW option of the NRCCellAlgoSwitch.LcsSwitch parameter	<i>Positioning</i>	None
SRS field strength-based positioning	SRS_RSRP_LOCATION_SW option of the NRCCellAlgoSwitch.LcsSwitch parameter	<i>Positioning</i>	None
Distributed massive MIMO	DM_MIMO_SERVICE_SWITCH option of the NRDUCellAlgoSwitch.DmMimoSwitch parameter	<i>Distributed Antenna Solutions (Low-Frequency TDD)</i>	Co-MM and distributed massive MIMO cannot be both enabled.
Fusion Cell	NRDUCell.NrDuCellNetworkingMode set to FUSION_CELL	<i>Fusion Cell (Low-Frequency TDD)</i>	Co-MM and Fusion Cell cannot be both enabled.
VMIMO in DAS-based indoor coverage	NRDUCellAlgoSwitch.VmimoSwitch	<i>Indoor Coverage</i>	Co-MM and VMIMO in DAS-based indoor coverage cannot be both enabled.

Function Name	Function Switch	Reference	Description
TRS rate matching optimization	TRS_RATEMATCH_OPT_SW option of the NRDUCellPdsch.RateMatchSwitch parameter	<i>Downlink Scheduling</i>	Co-MM and TRS rate matching optimization cannot be both enabled.
IAB deployment	NRDUCell.DeployPosition set to DEPLOY_IAB	None	None
gNodeB Adaptive Fast Processing Switch	GNB_ADAPT_FAST_PROCESS_SW option of the gNodeBParam.ServiceDiffProcSwitch parameter	None	Enhanced Co-MM (controlled by the INTRA_GNB_SU_COMM_ENH_SW option of the NRDUCellCollabServ.MultiCellMimoSwitch parameter) cannot be enabled when the gNodeB adaptive fast processing switch is on. This switch cannot be turned on when enhanced Co-MM is enabled.
Rate-matching-pattern-configuration-based PDCCH rate matching	PDCCH_RATEMATCH_SW option of the NRDUCellPdsch.RateMatchSwitch parameter	<i>Downlink Scheduling</i>	Rate-matching-pattern-configuration-based PDCCH rate matching must be disabled before coordination between serving and cooperating cells with different numbers of PDCCH symbols (controlled by the PDCCH_SYM_NUM_DIFF_COORD_SW option of the NRDUCellPdcch.PdcchAlgoEnhSwitch parameter) is enabled.

Function Name	Function Switch	Reference	Description
Time-domain interference avoidance	gNodeBAlgo.TimeIntrfAvoidSw set to SS54_1GP_TIME_INTRF_AVOID_SW	<i>Interference Avoidance</i>	When the NRDUCell.SlotStructure parameter is set to SS54 , the SS54_1GP_TIME_INTRF_AVOID_SW option of the gNodeBAlgo.TimeIntrfAvoidSw parameter and the INTRA_GNB_SU_COMM_SW option of the NRDUCellCollabServ.MultiCellMimoSwitch parameter cannot be both selected.
PDCCH Rate Matching Enhancement Switch	PDCCH_RATEMATCH_ENH_SW option of the NRDUCellPdsch.RateMatchingSwitch parameter	None	Co-MM and PDCCH Rate Matching Enhancement Switch cannot be both turned on.
Tunnel-based channel imbalance optimization	CSIRS_BEAM_POWER_BALANCE_SW option of the NRDUCellAlgoSwitch.BeamOptSwitch parameter	<i>Indoor Coverage</i>	Co-MM and tunnel-based channel imbalance optimization cannot be both enabled.
Distributed Antenna Weight Opt Switch	DISTRIBUTED_ANT_WEIGHT_OPT_SW option of the NRDUCellAlgoSwitch.BeamOptSwitch parameter	None	Co-MM and Distributed Antenna Weight Opt Switch cannot be both turned on.
Extended Cell Range (TDD)	NRDUCell.CellRadius set to a value greater than 14500	<i>Extended Cell Range</i>	Co-MM cannot be enabled when the NRDUCell.SlotStructure parameter is set to SS518 or SS818 .
Extended Cell Range of 90 km (TDD)	NRDUCell.CellRadius set to a value greater than 60000 and less than or equal to 90000	<i>Extended Cell Range</i>	Co-MM cannot be enabled when the NRDUCell.SlotStructure parameter is set to SS518 or SS818 . Therefore, Extended Cell Range of 90 km (TDD) cannot be enabled with Co-MM.

4.3.2.3 Function Impacts

Function Name	Function Switch	Reference	Description
Downlink intra-band CA	INTRA_BAND_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	When CA is enabled, Co-MM cannot be performed for CA UEs in their SCells.
Downlink intra-FR inter-band CA	INTRA_FR_INTER_BAND_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	When CA is enabled, Co-MM cannot be performed for CA UEs in their SCells.
Inter-gNodeB CA	INTER_GNODEB_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>CA Experience Improvement</i>	When inter-gNodeB CA is enabled, Co-MM cannot be performed for CA UEs in their SCells.
SRS remote interference avoidance	SRS_RIM_INTER_AVOID_SW option of the NRDUCellSrs.SrsAlgoExtSwitch parameter	<i>Remote Interference Management (Low-Frequency TDD)</i>	When SRS remote interference avoidance is enabled in the serving cell but disabled in the cooperating cell, Co-MM does not take effect.
RF channel intelligent shutdown	RF_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When RF channel intelligent shutdown takes effect in a cell, Co-MM stops taking effect for UEs involved in Co-MM in the cell and will not take effect for UEs that are newly admitted to this cell. After the cell exits the RF channel intelligent shutdown state, Co-MM takes effect again.

Function Name	Function Switch	Reference	Description
LNR coordinated channel shutdown	MULTI_RAT_RF_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Multi-RAT Coordinated Energy Saving</i>	When LNR coordinated channel shutdown takes effect in a cell, Co-MM stops taking effect for UEs involved in Co-MM in the cell and will not take effect for UEs that are newly admitted to this cell. After the cell exits the LNR coordinated channel shutdown state, Co-MM takes effect again.
PDSCH MU-MIMO	DL_MU_MIMO_SW option of the NRDUCellAlgoSwitch.MuMimoSwitch parameter	<i>MIMO (TDD)</i>	Co-MM UEs cannot be paired for PDSCH MU-MIMO.
Power saving BWP	BWP2_SWITCH option of the NRDUCellUePwrSaving.BwpPwrSavingSw parameter	<i>UE Power Saving</i>	If the COOR_UE_PWR_SAVE_PRI_ADJ_SW option of the NRDUCellCollabServ.CommAlgoSwitch parameter is selected, the downlink throughput of Co-MM UEs increases. However, the proportion of time in which UEs stay in the power saving state (that is, UEs work on BWP2) may decrease, the value of the N.Bwp2.Dur.Total counter decreases, and the UE power saving effect may deteriorate. The SRS SINR measurement value may increase (indicated by a decrease in the proportion of SINR measurement values that fall into small-index ranges).

Function Name	Function Switch	Reference	Description
Coordinated interference management	INTRA_GNB_DL_CS_CBF_SW and INTER_GNB_DL_CS_CBF_SW options of the NRDUCellAlgoSwitch.CompSwitch parameter	<i>Coordinated Interference Management (Low-Frequency TDD)</i>	Coordinated interference management is not performed for UEs involved in Co-MM.
DRX	BASIC_DRX_SW option of the NRDUCellUePwrSaving.NrDuCellDrxAlgoSwitch parameter	<i>DRX</i>	- After DRX is enabled, UEs are less likely to be identified as Co-MM UEs. - After DRX is enabled, if exit of UEs performing full-buffer services from DRX (controlled by the FULL_BUFFER_UE_EXIT_DRX_SW option of the NRDUCellUePwrSaving.NrDuCellDrxAlgoSwitch parameter) is enabled, DRX (sleep or active time) does not take effect for Co-MM UEs. If exit of UEs performing full-buffer services from DRX is disabled, Co-MM UEs cannot enter the DRX sleep time because the aperiodic CSI-RS for CM is used.

Function Name	Function Switch	Reference	Description
Operator-specific limitation on the maximum number of available RBs	MAX_RB_LIMIT_SW option of the NRDUCellAlgoSwitch.RanSharingAlgoSwitch parameter	<i>Multi-Operator Sharing</i>	When operator-specific limitation on the maximum number of available RBs and Co-MM are both enabled: - The RBs available to Co-MM UEs are subject to the maximum number of available RBs. - If the NRDUCellRes.DIRbMaxRatio parameter is set to a small value (for example, less than 50%) for a secondary operator, UEs of this operator may not be identified as Co-MM UEs.
PUCCH RB adaptation	PUCCH_RBRES_ADAPTIVE_SWITCH option of the NRDUCellPucch.PucchAlgoSwitch parameter	<i>Channel Management</i>	When Co-MM takes effect, it is recommended that PUCCH RB adaptation be enabled. Enabling this function protects non-Co-MM UEs against service drops that occur due to resource preemption by Co-MM UEs.
CSI reporting period adaptation	CSI_REPORT_PERIOD_ADAPT_SWITCH option of the NRDUCellPucch.CsiResoureAlgoSwitch parameter	<i>Channel Management</i>	When Co-MM takes effect, it is recommended that CSI reporting period adaptation be enabled. Enabling this function protects non-Co-MM UEs against service drops that occur due to resource preemption by Co-MM UEs.

Function Name	Function Switch	Reference	Description
Full-buffer UE-specific CSI-RS for CM	NRDUCellCsirs.CsiUserPeriod	<i>Channel Management</i>	When the NRDUCellComp.JtCsirsTxMode parameter is set to PERIODIC_TD_MODE_ABN or PERIODIC_TD_MODE_ALL , Co-MM takes effect only if the NRDUCellCsirs.CsiUserPeriod parameter is set to SLOT20 , SLOT40 , or SLOT80 .
JT CSI-RS Transmit Mode	NRDUCellComp.JtCsirsTxMode	None	
Network slicing	Multiple parameters are required for network slicing deployment. For details, see <i>Network Slicing</i> .	<i>Network Slicing</i>	The network slices configured for a cooperating cell must include the UE's activated network slices.

Function Name	Function Switch	Reference	Description
Rate-matching-pattern-configuration-based PDCCH rate matching	PDCCH_RATEMATCH_SW option of the NRDUCellPdsch.RateMatchSwitch parameter	<i>Downlink Scheduling</i>	- When the JT_PDCCH_RM_COORD_SW option of the NRDUCellComp.DlCompAlgoSwitch parameter is selected, rate-matching-pattern-configuration-based PDCCH rate matching and Co-MM can simultaneously take effect if both are enabled. - When the JT_PDCCH_RM_COORD_SW option of the NRDUCellComp.DlCompAlgoSwitch parameter is deselected, either rate-matching-pattern-configuration-based PDCCH rate matching or Co-MM (not both; adaptively selected) takes effect on UEs in overlapping areas if both are enabled.

Function Name	Function Switch	Reference	Description
SSB rate matching	SSB_RATEMATCH_SW option of the NRDUCellPdsch.RateMatchSwitch parameter	<i>Downlink Scheduling</i>	Co-MM is less likely to take effect if SSB rate matching, rate-matching-pattern-configuration-based PDCCH rate matching, and collaboration between joint transmission and PDCCH rate matching are enabled for both the serving and cooperating cells that use different SSB patterns or PDCCH rate matching patterns.
Rate-matching-pattern-configuration-based PDCCH rate matching	PDCCH_RATEMATCH_SW option of the NRDUCellPdsch.RateMatchSwitch parameter	<i>Downlink Scheduling</i>	
Collaboration between joint transmission and PDCCH rate matching	JT_PDCCH_RM_COORD_SW option of the NRDUCellComp.DlCompAlgoSwitch parameter	None	To increase this probability, it is recommended that the JT_SSB_RM_COORD_SW option of the NRDUCellComp.DlCompAlgoExtSwitch parameter be selected to enable coordination between joint transmission and SSB rate matching.
External interference avoidance	UL_UE_LEVEL_INTERFERENCE_AVOIDANCE option of the NRDUCellIntrflIdent.IntrfOptimizationMethod parameter and the NRDUCellIntrflIdent.IntrflIdentificationMethod parameter set to UL_MANUAL or UL_AUTO	<i>Interference Avoidance</i>	After external interference avoidance is enabled, there is no SRS transmission in the interference range. As a result, the probability that Co-MM takes effect decreases.

Function Name	Function Switch	Reference	Description
Frequency offset measurement optimization based on interference suppression	INTRF_FREQOF FSET_OPT_SW option of the NRDUCellPusch.PuschMeasurement parameter	<i>Interference Avoidance</i>	After frequency offset measurement optimization based on interference suppression takes effect, the average downlink cell throughput may increase in Co-MM scenarios.
Power saving scheduling	POWER_SAVING_SCHEDULE_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When Co-MM and power saving scheduling are both enabled, the power saving gains may be relatively low.
Intelligent carrier shutdown	INTRA_GNB_MULTICARRIER_SHUTDOWN_SW or INTER_GNB_MULTICARRIER_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When intelligent carrier shutdown takes effect, the probability that Co-MM takes effect decreases.
NR inter-carrier dynamic power sharing	NR_DYN_POWER_SHARING_SW option of the NRDUCellAlgoSwitch.DynPowerSharingSwitch parameter	<i>NR Inter-Carrier Dynamic Power Allocation</i>	When Co-MM and NR inter-carrier dynamic power sharing are both enabled, the gains provided by NR inter-carrier dynamic power sharing may decrease.
LTE TDD and NR Flash Dynamic Power Sharing	LTE_NR_DYN_POWER_SHARING_SW option of the NRDUCellAlgoSwitch.DynPowerSharingSwitch parameter	<i>LTE and NR Power Allocation</i>	When Co-MM and LTE TDD and NR Flash Dynamic Power Sharing are both enabled, the power sharing gains on the NR side may decrease.

Function Name	Function Switch	Reference	Description
L/NR Inter-band Deep Coordination	LNR_INTER_BAND_DEEP_COORD_SW option of the NRDUCellResS hrCfg.DynPwrShrAlgoSwitch parameter	<i>LTE and NR Power Allocation</i>	When L/NR Inter-band Deep Coordination and Co-MM are both enabled, the power sharing gains on the NR side may decrease.
Cell-level MRO	MRO_SCENARIO_IDENT_SW option of the gNB Mro.ScenarioIdentifySwitch parameter and the INTRA_FREQ_MRO_SW option of the NRCCellMro.MroSwitch parameter	<i>MRO</i>	If Co-MM and intra-frequency MRO are both enabled, event A3 reporting in Co-MM is affected. This further affects identification of UEs in overlapping areas for Co-MM.
UE-level MRO	UE_MRO_SWITCH option of the NRCCellMro.MroSwitch parameter	<i>MRO</i>	If Co-MM and UE-level MRO are both enabled and Co-MM takes effect for ping-pong UEs, event A3 reporting in Co-MM is affected. This further affects identification of UEs in overlapping areas for Co-MM.
Downlink type 1 scheduling for slice groups with the TTI-based control policy	DL_TYPE1_SCHED_SW option of the NRDUCellRes.SchedulingPolicySwitch parameter	<i>Network Slicing</i>	When downlink type 1 scheduling for slice groups with the TTI-based control policy takes effect, Co-MM does not take effect for UEs in the slice groups.
PDCCH skipping	PDCCH_SKIP_SW option of the NRDUCellUePwrSaving.SchPwrSavingSw parameter	<i>UE Power Saving</i>	When Co-MM and PDCCH skipping are both enabled, PDCCH skipping does not take effect on Co-MM UEs.

Function Name	Function Switch	Reference	Description
CCE resource optimization	CCE_RESOURCE_OPT_SW option of the NRDUCellPdcch.PdcchAlgorithmSwitch parameter	<i>MIMO (TDD)</i>	When CCE resource optimization takes effect, the probability that Co-MM takes effect may decrease. To raise the probability, it is recommended that coordination between serving and cooperating cells with different numbers of PDCCH symbols (controlled by the PDCCH_SYM_NUM_DIFF_COORD_SW option of the NRDUCellPdcch.PdcchAlgorithmSwitch parameter) be enabled.
Clock out-of-synchronization detection	CLK_DETECT_SW option of the gNodeBParam.ClkOutOfSyncDetectionSwitch parameter	<i>Synchronization</i>	Clock out-of-synchronization detection may affect the downlink user-perceived rates of Co-MM UEs. It is recommended that the INTRF_STAT_ALGO_ENH_SW option of the gNBCLKResMode.ClkOutOfSyncAlgorithmSw parameter be selected to reduce the number of times clock out-of-synchronization detection is performed in the case of non-continuous external interference.

Function Name	Function Switch	Reference	Description
PDCCH symbol number adaptation	UE_PDCCH_SYM_NUM_ADAPT_SW option of the NRDUCellPdcch.PdcchAlgoEnhSwitch parameter	<i>Channel Management</i>	Co-MM requires that the cooperating and serving cells have the same time-domain resources for PDSCH data transmission. Therefore, the number of symbols occupied by the PDCCH in the cooperating cell must be less than or equal to that in the serving cell. When PDCCH symbol number adaptation is enabled, the number of symbols occupied by the PDCCH may differ between the serving and cooperating cells. • With the PDCCH_SYM_NUM_DIFF_COORD_SW option of the NRDUCellPdcch.PdcchAlgoEnhSwitch parameter deselected, Co-MM can take effect only if the number of symbols occupied by the PDCCH in the cooperating cell is the same as that in the serving cell. • With the PDCCH_SYM_NUM_DIFF_COORD_SW option of the NRDUCellPdcch.PdcchAlgoEnhSwitch parameter selected, Co-MM can take effect if the number of symbols occupied by the PDCCH in the cooperating cell is either less than or equal to that in the serving cell.

Function Name	Function Switch	Reference	Description
LTE TDD and NR spectrum sharing	LTE_NR_TDD_S PCT_SHR_SW option of the NRDUCellAlgo Switch.Spectru mCloudSwitch parameter	None	When LTE TDD and NR spectrum sharing takes effect, the number of available downlink RBs in the NR cell dynamically changes, affecting the probability that Co-MM takes effect.
Downlink 8-layer transmission for a single FWA user	DL_SU_EIGHT_ LAYER_SW option of the NRDUCellDIRa nk.DIRankSela lgoSw parameter	<i>FWA</i>	If downlink 8-layer transmission for a single FWA user has taken effect, Co-MM does not take effect for the user.
Uplink 4-layer transmission for a single FWA user	UL_SU_FOUR_L AYER_SW option of the NRDUCellUIRa nk.UIRankAlgo Sw parameter	<i>FWA</i>	If uplink 4-layer transmission for a single FWA user has taken effect, Co-MM does not take effect for the user.

Function Name	Function Switch	Reference	Description
CA Performance Boost Based on Multi-dimensional Optimization	CA_PRFM_BOOST_BSD_ON_MTD_OPT_SW option of the NRDUCellCarrierMgmt.CaEnhancedAlgoExtSwitch parameter	<i>Multi-Frequency Smart Aggregation</i>	- CA Performance Boost Based on Multi-dimensional Optimization does not take effect for UEs for which Co-MM has taken effect. - On a given gNodeB, if CA Performance Boost Based on Multi-dimensional Optimization is enabled only for some of the cells for which intra-base-station Co-MM is enabled, the number of cells in which CA Performance Boost Based on Multi-dimensional Optimization takes effect may be less than the number of cells for which this function is enabled. Therefore, it is recommended that CA Performance Boost Based on Multi-dimensional Optimization be enabled for all cells under the gNodeB for which intra-base-station Co-MM is enabled.
LTE and NR multi-band power adaptive scheduling	LNR_MULTI_BAND_PWR_ADPT_SCH_SW option of the NRDUCellAlgoSwitch.DynPowerSharingSwitch parameter	<i>LTE and NR Power Allocation</i>	When LTE and NR multi-band power adaptive scheduling and Co-MM are both enabled, there is a possibility that the benefits of power adaptive scheduling decrease on the NR side.

Function Name	Function Switch	Reference	Description
NR multi-band power adaptive scheduling	NR_MULTI_BAND_PWR_ADAPT_SCH_SW option of the NRDUCellAlgoSwitch.DynPowerSharingSwitch parameter	<i>NR Inter-Carrier Dynamic Power Allocation</i>	When Co-MM and NR multi-band power adaptive scheduling are both enabled, there is a possibility that the benefits of the latter decrease.
SSSG switching	UE_PWR_SAVING_ENHANCE_SW and SSSG_SWITCHING_SW options of the NRDUCellUePowerSaving.UePowerSavingSwitch parameter	<i>UE Power Saving</i>	UEs with Co-MM in effect will not switch from a dense search space set group (SSSG) to a sparse one. For Co-MM to take effect for a UE with a sparse SSSG in effect, the UE must switch to a dense SSSG.
CCE partial symbol adaptation	CCE_PARTIAL_SYM_ADAPT_SW option of the NRDUCellPdcch.PdcchAlgoSwitch parameter	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	When Co-MM takes effect, the benefits of CCE partial symbol adaptation will decrease.
Inter-cell interference avoidance	INTER_CELL_INTERF_AVOID_SW option of the NRDUCellCollabServ.MultiCellMimoSwitch parameter	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	Inter-cell interference avoidance does not apply to UEs for which Co-MM is in effect.
Adaptive filtering for SRS channel estimation	CHN_EST_ADAPT_FLTR_SW option of the NRDUCellSrs.SrsDetectionAlgoSwitch parameter	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	Adaptive filtering for SRS channel estimation may yield fewer benefits when it is enabled together with enhanced Co-MM.
Self-fusion weights	DL_SELF_FUSION_WEIGHT_SW option of the NRDUCellPdschPrecode.PrecodingAlgoSwitch parameter	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	The self-fusion weights function will yield fewer benefits when it is enabled together with enhanced Co-MM.

Function Name	Function Switch	Reference	Description
Downlink precise RB estimation	PRECISE_DL_RB_ESTIMATE_SW option of the NRDUCellDLSch.DLSchAlgoSwitch parameter	<i>Downlink Scheduling</i>	Co-MM and enhanced Co-MM may yield more benefits when downlink precise RB estimation is enabled.
RedCap SSSG switching	REDCAP_SSSG_SWITCHING_SW option of the NRDUCellUePwrSaving.UePowerSavingSwitch parameter REDCAP_SSSG_SWITCH_EXT_SW option of the NRDUCellUePwrSaving.UePowerSavingSwitch parameter	<i>RedCap</i>	UEs with Co-MM in effect will not switch from a dense SSSG to a sparse one. For Co-MM to take effect for a UE with a sparse SSSG in effect, the UE must switch to a dense SSSG.
Time-domain power saving BWP (low-frequency TDD)	TDM_BWP2_SWITCH option of the NRDUCellUePwrSaving.BwpPwrSavingSw parameter	<i>UE Power Saving</i>	If the COOR_UE_PWR_SAVE_PRI_ADJ_SW option of the NRDUCellCollabServ.CoMmAlgoSwitch parameter is selected, the downlink throughput of Co-MM UEs increases. However, the proportion of time in which UEs work on BWP2 may decrease, the value of the N.Bwp2.Dur.Total counter decreases, and the UE power saving effect may deteriorate. The SRS SINR measurement value may increase (indicated by a decrease in the proportion of SINR measurement values that fall into small-index ranges).

Function Name	Function Switch	Reference	Description
Proactive avoidance of remote interference	SRS_RIM_INTRF_AVOID_SW and SRS_INTRF_COORD_S_SLOT_SW options of the NRDUCellSrs.SrsAlgoExtSwitch parameter	<i>Remote Interference Management (Low-Frequency TDD)</i>	If the SRS_RIM_INTRF_AVOID_SW option of the NRDUCellSrs.SrsAlgoExtSwitch parameter is selected for the serving cell but deselected for the cooperating cell, Co-MM does not take effect. If this option is deselected for the serving cell or selected for both the serving and cooperating cells, Co-MM can take effect.
RF module fast deep dormancy	FAST_DEEP_DORM_SW option of the RRUEXTINFO.RUDOROPTFEATSW parameter	<i>Multi-RAT Coordinated Energy Saving</i>	After RF module fast deep dormancy is enabled, the probability of Co-MM taking effect decreases.

4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

Boards

All NR-capable main control boards and NR TDD-capable baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

SRS-based weight enhancement for joint transmission requires UBBPg series or later baseband processing units in 4T4R and 8T8R scenarios. In other scenarios, all NR TDD-capable baseband processing units support this function.

RF Modules

The following RF modules are compatible with this function:

- 32T32R or higher: NR TDD-capable RF modules that comply with the eCPRI protocol and work in band n38, n40, n41, n77, n78, or n79
- 8T8R or lower:
 - When the start base station version that supports board software mapping is earlier than V100R020C00: NR TDD-capable RF modules that work in band n38, n40, n41, n77, n78, or n79, including the RRU5258, RRU5818, RRU5818w (02314FXB), RRU5836E (02312LUK), RRU5254, RRU5810, and AAU5243
 - When the start base station version that supports board software mapping is V100R020C00 or later: all NR TDD-capable 8T8R and 4T4R RF modules

For details about the start base station version that supports board software mapping, see *3900 & 5900 Series Base Station Software and Hardware Compatibility Description* in *3900 & 5900 Series Base Station Product Documentation*. For details about the supported RATs and TX/RX channel configurations of each RF module, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

4.3.4 Cells

The **NRDUCell.DuplexMode** parameter must be set to **CELL_TDD**.

The **NRDUCell.FrequencyBand** parameter must be set to **N38, N40, N41, N77, N78, or N79**.

4.3.5 Others

Co-MM UEs must be capable of antenna selection and support 1T4R, 2T4R, or 4T4R, as indicated by the supportedSRS-TxPortSwitch field being t1r4, t2r4, or t4r4 in the *UE capability information elements* IE. For details, see section 6.3.3 "UE capability information elements" in 3GPP TS 38.331 V17.1.0.

The gains brought by Co-MM are highly dependent on the cluster ID plan and the quality of overlapping area optimization. To obtain the desired gains, contact Huawei professional optimization service engineers for operation instructions.

Co-MM involves over-the-air channel calibration between two TRPs. In scenarios where the inter-site distance is short and a cluster includes many cells, you can manually configure the signal response relationship between AAUs/RRUs and the calibration filtering coefficient to achieve the desired gains. The manual configuration helps improve the stability of over-the-air channel calibration for Co-MM clusters. For detailed operations, contact Huawei professional optimization service engineers.

If the AAU adopts inter-BBU SDR networking, the corresponding BBU must meet the configuration requirements for this networking. For details about the configuration requirements for inter-BBU SDR, see *BBU Interconnection*.

4.4 Operation and Maintenance

4.4.1 Data Configuration

4.4.1.1 Data Preparation

Table 4-7 describes the parameters used for function activation.

Table 4-7 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Multi-Cell MIMO Switch	NRDUCellCollab Serv.<i>MultiCellMimoSwitch</i>	INTRA_GNB_S U_COMM_SW	Co-MM requires that this option be selected in the serving and neighboring cells to be engaged in collaboration.
Cluster ID	gNBCluster.<i>ClusterId</i>	None	Set this parameter based on the network plan. It is recommended that the ClusterId settings of adjacent clusters be staggered in a mod-4 manner (based on the remainder of the division of ClusterId by 4). If this staggering condition is not met, channel calibration will fail.
Cluster Type	gNBCluster.<i>ClusterType</i>	None	Set this parameter to MULTI_CELL_MIMO .
Cluster ID	gNBMimoClusterCell.<i>ClusterId</i>	None	Set this parameter based on the network plan.
Mobile Country Code	gNBMimoClusterCell.<i>Mcc</i>	None	Set this parameter based on the network plan.
Mobile Network Code	gNBMimoClusterCell.<i>Mnc</i>	None	Set this parameter based on the network plan.
gNodeB ID	gNBMimoClusterCell.<i>gNodeBId</i>	None	Set this parameter based on the network plan.
Cell ID	gNBMimoClusterCell.<i>CellId</i>	None	Set this parameter based on the network plan.
Calibration Robustness Switch	gNBCluster.<i>CalibRobustSw</i>	CALIB_FILTER_SW	Set this parameter based on the network plan.
Calibration Robustness Switch	gNBCluster.<i>CalibRobustSw</i>	CALIB_CELL_ROBUST_SW	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Option	Setting Notes
Calibration Filter Coefficient	gNBCluster.CalibFilterCoeff	None	Set this parameter based on the network plan.
CoMP Switch	NRDUCellAlgoSwitch.CompSwitch	INTRA_GNB_OVERLAPPING_MEAS_SW	It is recommended that this option be selected one week before Co-MM is enabled to observe the throughput changes before and after Co-MM is enabled.
CoMP Switch	NRDUCellAlgoSwitch.CompSwitch	INTER_GNB_OVERLAPPING_MEAS_SW	It is recommended that this option be selected one week before Co-MM is enabled to observe the throughput changes before and after Co-MM is enabled.
DL CoMP RSRP Offset	NRCCellComp.DLCompRsrpOffset	None	Retain the default value.
Overlapping RSRP Offset	NRCCellComp.OverlappingRsrpOffset	None	It is recommended that this parameter be set to the value of NRCCellComp.DLCompRsrpOffset .
Hysteresis	NRCCellComp.Hysteresis	None	Retain the default value.
Time to Trigger	NRCCellComp.TimeToTrigger	None	Retain the default value.
Cooperating Cell RSRP Threshold	NRDUCellCollabServ.CoCellRsrpThld	None	Retain the default value.
SRS RSRP Difference Threshold	NRDUCellCollabServ.SrsRsrpDiffThld	None	Retain the default value.
Cooperating Cell PreSINR Threshold	NRDUCellCollabServ.CoopCellPreSnrThld	None	Retain the default value.
DL CoMP Coordinating Cell PRB Threshold	NRDUCellComp.DLCompCoCellPrbThld	None	Retain the default value.

Parameter Name	Parameter ID	Option	Setting Notes
JT Quick Rollback Spectral Efficiency Thld	NRDUCellComp. <i>JtQuickRollbackSpctEffThld</i>	None	Retain the default value.
JT Delay Compensation Threshold	NRDUCellComp. <i>JtDelayCompensationThld</i>	None	Retain the default value.
DL CoMP Algorithm Switch	NRDUCellComp. <i>DlCompAlgoSwitch</i>	JT_JUDGE_OPT_SW	You are advised to select this option.
DL CoMP Algorithm Switch	NRDUCellComp. <i>DlCompAlgoSwitch</i>	JT_SRS_WEIGHT_ENH_SW	You are advised to select this option.
DL CoMP Algorithm Switch	NRDUCellComp. <i>DlCompAlgoSwitch</i>	JT_RANK_ADAPT_SW	You are advised to select this option.
Downlink PDSCH Algorithm Switch	NRDUCellPdsch. <i>DlPdschAlgoSwitch</i>	RMSI_RES_RSV_OPT_SW	You are advised to select this option.
SU-MIMO Power Control Protection Threshold	NRDUCellDlAmc. <i>SuMimoPwrCtrlProtectThld</i>	None	It is recommended that this parameter be set to 35 when the signal quality differs greatly between streams.
CoMM Algorithm Switch	NRDUCellCollab <i>Serv.CommAlgoSwitch</i>	WEIGHT_OPT_SW	It is recommended that this option be selected in the case of abnormal cross-correlation.
CoMM Algorithm Switch	NRDUCellCollab <i>Serv.CommAlgoSwitch</i>	COMM_PHASE_COMPENSATION_SW	It is recommended that this option be selected to increase the downlink throughput of Co-MM UEs in mobility.
Co-MM Calibration Factor	gNBCluster. <i>CoMMCalibFactor</i>	None	Retain the default value.
Multi-Cell MIMO Switch	NRDUCellCollab <i>Serv.MultiCellMimoSwitch</i>	INTRA_GNB_SU_COMM_ENH_SW	You are advised to select this option.

Parameter Name	Parameter ID	Option	Setting Notes
CoMM Algorithm Switch	NRDUCellCollab Serv.<i>CommAlgo</i> Switch	FIRST_PACKET_ SCH_DELAY_ SW	It is recommended that this option be selected to increase the downlink throughput of Co-MM UEs.
PDCCH Algorithm Enhancement Switch	NRDUCellPdcch. PdcchAlgoEnhS witch	PDCCH_SYM_ NUM_DIFF_CO ORD_SW	You are advised to select this option.
DL CoMP Algorithm Extension Switch	NRDUCellComp. DLCompAlgoExt Switch	JT_SSB_RM_CO ORD_SW	When SSB rate matching is enabled for both the serving and cooperating cells, it is recommended that this option be selected to increase the probability that Co-MM takes effect.
Multi-carrier Calibration Factor	gNodeBParam. MulticarrierCali bFactor	None	This parameter specifies the multi-carrier calibration factor for Co-MM to take effect on one or more carriers. In single-carrier scenarios, it is recommended that this parameter be set to 1. In multi-carrier scenarios, it is recommended that this parameter be set to the number of carriers. Co-MM can take effect on three or four carriers on a dual-band AAU only if the AAU model is one of the following: - AAU5368md (2300 MHz+3500 MHz) - AAU5368md (2600 MHz+3600 MHz) When modules of a preceding model have three or four carriers configured on each and provide contiguous coverage, it is recommended that this parameter be set to the number of carriers.

Parameter Name	Parameter ID	Option	Setting Notes
CoMM Algorithm Switch	NRDUCellCollab Serv.<i>CommAlgo</i> Switch	COOR_UE_PW R_SAVE_PRI_A DJ_SW	It is recommended that this option be selected to increase the downlink throughput of Co-MM UEs. However, when both Co-MM and power saving BWP are enabled, the UE power saving effect may deteriorate.
CoMP Optimization Switch	NRDUCellComp. CompOptSwitc h	COOP_CELL_A LLOC_OPT_SW	It is recommended that this option be selected to increase the probability of Co-MM taking effect.

4.4.1.2 Using MML Commands

Before using MML commands, refer to [4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

To compare the throughput before and after Co-MM is enabled, you are advised to set parameters as follows one week before Co-MM is enabled. This ensures that sufficient statistics about UE throughput in overlapping areas can be collected for comparison.

```
//Enabling intra-base-station overlapping area measurement
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,
CompSwitch=INTRA_GNB_OVERLAPPING_MEAS_SW-1&INTER_GNB_OVERLAPPING_MEAS_SW-0;
MOD NRDUCELLALGOSWITCH: NrDuCellId=1,
CompSwitch=INTRA_GNB_OVERLAPPING_MEAS_SW-1&INTER_GNB_OVERLAPPING_MEAS_SW-0;
//Setting parameters related to intra-base-station overlapping area measurement
MOD NRCELLCOMP: NrCellId=0, OverlappingRsrpOffset=-18, DlCompRsrpOffset=-18,
TimeToTrigger=320MS, Hysteresis=2;
MOD NRCELLCOMP: NrCellId=1, OverlappingRsrpOffset=-18, DlCompRsrpOffset=-18,
TimeToTrigger=320MS, Hysteresis=2;
```

Activation Command Examples

```
//Adding a gNodeB cluster
ADD GNBCLUSTER: ClusterId=0, ClusterType=MULTI_CELL_MIMO, CoMMCalibFactor=4;
//Adding gNodeB MIMO cluster cell configurations, with Mcc and Mnc specified based on site conditions
ADD GNBMMIMOCCLUSTERCELL: ClusterId=0, Mcc="123", Mnc="45", gNodeBId=0, CellId=0;
ADD GNBMMIMOCCLUSTERCELL: ClusterId=0, Mcc="123", Mnc="45", gNodeBId=0, CellId=1;
//Enabling Co-MM
MOD NRDUCELLCOLLABSERV: NrDuCellId=0, MultiCellMimoSwitch=INTRA_GNB_SU_COMM_SW-1;
MOD NRDUCELLCOLLABSERV: NrDuCellId=1, MultiCellMimoSwitch=INTRA_GNB_SU_COMM_SW-1;
//Turning on calibration robustness switches and setting the calibration filtering coefficient
```

```

MOD GNBCLUSTER: ClusterId=0, CalibRobustSw=CALIB_FILTER_SW-1&CALIB_CELL_ROBUST_SW-1,
CalibFilterCoeff=90;
//Setting the CoCellRsrpThld parameter
MOD NRDUCELLCOLLABSERV: NrDuCellId=0, CoCellRsrpThld=-90;
MOD NRDUCELLCOLLABSERV: NrDuCellId=1, CoCellRsrpThld=-90;
//Setting the SrsRsrpDiffThld parameter
MOD NRDUCELLCOLLABSERV: NrDuCellId=0, SrsRsrpDiffThld=-20;
MOD NRDUCELLCOLLABSERV: NrDuCellId=1, SrsRsrpDiffThld=-20;
//Setting the CoopCellPreSinrThld parameter
MOD NRDUCELLCOLLABSERV: NrDuCellId=0, CoopCellPreSinrThld=-20;
MOD NRDUCELLCOLLABSERV: NrDuCellId=1, CoopCellPreSinrThld=-20;
//Setting the DlCompCoCellPrbThld parameter
MOD NRDUCELLCOMP: NrDuCellId=0, DlCompCoCellPrbThld=30;
MOD NRDUCELLCOMP: NrDuCellId=1, DlCompCoCellPrbThld=30;
//Setting the JtQuickRollbackSpctEffThld parameter
MOD NRDUCELLCOMP: NrDuCellId=0, JtQuickRollbackSpctEffThld=95;
MOD NRDUCELLCOMP: NrDuCellId=1, JtQuickRollbackSpctEffThld=95;
//Setting the JtDelayCompensationThld parameter
MOD NRDUCELLCOMP: NrDuCellId=0, JtDelayCompensationThld=15;
MOD NRDUCELLCOMP: NrDuCellId=1, JtDelayCompensationThld=15;
//Enabling joint transmission evaluation optimization
MOD NRDUCELLCOMP: NrDuCellId=0, DlCompAlgoSwitch=JT_JUDGE_OPT_SW-1;
MOD NRDUCELLCOMP: NrDuCellId=1, DlCompAlgoSwitch=JT_JUDGE_OPT_SW-1;
//Enabling SRS-based weight enhancement for joint transmission
MOD NRDUCELLCOMP: NrDuCellId=0, DlCompAlgoSwitch=JT_SRS_WEIGHT_ENH_SW-1;
MOD NRDUCELLCOMP: NrDuCellId=1, DlCompAlgoSwitch=JT_SRS_WEIGHT_ENH_SW-1;
//Enabling rank adaptation for joint transmission
MOD NRDUCELLCOMP: NrDuCellId=0, DlCompAlgoSwitch=JT_RANK_ADAPT_SW-1;
MOD NRDUCELLCOMP: NrDuCellId=1, DlCompAlgoSwitch=JT_RANK_ADAPT_SW-1;
//Enabling RMSI resource reservation optimization
MOD NRDUCELLPDSCH: NrDuCellId=0, DlPdschAlgoSwitch=RMSI_RES_RSV_OPT_SW-1;
MOD NRDUCELLPDSCH: NrDuCellId=1, DlPdschAlgoSwitch=RMSI_RES_RSV_OPT_SW-1;
//Setting the SuMimoPwrCtrlProtectThld parameter
MOD NRDUCELLDLAMC: NrDuCellId=0, SuMimoPwrCtrlProtectThld=35;
MOD NRDUCELLDLAMC: NrDuCellId=1, SuMimoPwrCtrlProtectThld=35;
//Enabling weight optimization for Co-MM
MOD NRDUCELLCOLLABSERV: NrDuCellId=0, CommAlgoSwitch=WEIGHT_OPT_SW-1;
MOD NRDUCELLCOLLABSERV: NrDuCellId=1, CommAlgoSwitch=WEIGHT_OPT_SW-1;
//Enabling phase compensation for Co-MM
MOD NRDUCELLCOLLABSERV: NrDuCellId=0, CommAlgoSwitch=COMM_PHASE_COMPENSATION_SW-1;
MOD NRDUCELLCOLLABSERV: NrDuCellId=1, CommAlgoSwitch=COMM_PHASE_COMPENSATION_SW-1;
//Enabling enhanced Co-MM
MOD NRDUCELLCOLLABSERV: NrDuCellId=0, MultiCellMimoSwitch=INTRA_GNB_SU_COMM_ENH_SW-1;
MOD NRDUCELLCOLLABSERV: NrDuCellId=1, MultiCellMimoSwitch=INTRA_GNB_SU_COMM_ENH_SW-1;
//Enabling delayed first-packet scheduling
MOD NRDUCELLCOLLABSERV: NrDuCellId=0, CommAlgoSwitch=FIRST_PACKET_SCH_DELAY_SW-1;
MOD NRDUCELLCOLLABSERV: NrDuCellId=1, CommAlgoSwitch=FIRST_PACKET_SCH_DELAY_SW-1;
//Enabling coordination between serving and cooperating cells with different numbers of PDCCCH symbols
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoEnhSwitch=PDCCCH_SYM_NUM_DIFF_COORD_SW-1;
MOD NRDUCELLPDCCH: NrDuCellId=1, PdcchAlgoEnhSwitch=PDCCCH_SYM_NUM_DIFF_COORD_SW-1;
//Enabling coordination between joint transmission and SSB rate matching when SSB rate matching is
enabled for both the serving and cooperating cells
MOD NRDUCELLCOMP: NrDuCellId=0, DlCompAlgoExtSwitch=JT_SSB_RM_COORD_SW-1;
MOD NRDUCELLCOMP: NrDuCellId=1, DlCompAlgoExtSwitch=JT_SSB_RM_COORD_SW-1;
//Setting the multi-carrier calibration factor (using single-carrier networking as an example)
MOD GNODEBPARAM: MulticarrierCalibFactor=1;
//Enabling priority adjustment for coordination and UE power saving
MOD NRDUCELLCOLLABSERV: NrDuCellId=0, CommAlgoSwitch=COOR_UE_PWR_SAVE_PRI_ADJ_SW-1;
MOD NRDUCELLCOLLABSERV: NrDuCellId=1, CommAlgoSwitch=COOR_UE_PWR_SAVE_PRI_ADJ_SW-1;
//Enabling optimized cooperating cell selection
MOD NRDUCELLCOMP: NrDuCellId=0, CompOptSwitch=COOP_CELL_ALLOC_OPT_SW-1;
MOD NRDUCELLCOMP: NrDuCellId=1, CompOptSwitch=COOP_CELL_ALLOC_OPT_SW-1;

```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Turning off the switches for overlapping area measurement to reduce signaling exchanges over the air
//interface (recommended when the throughput of UEs in the overlapping areas does not need to be
//observed)
MOD NRDUCELLALGOSWITCH: NrDuCellId=0,
CompSwitch=INTRA_GNB_OVERLAPPING_MEAS_SW-0&INTER_GNB_OVERLAPPING_MEAS_SW-0;
MOD NRDUCELLALGOSWITCH: NrDuCellId=1,
CompSwitch=INTRA_GNB_OVERLAPPING_MEAS_SW-0&INTER_GNB_OVERLAPPING_MEAS_SW-0;
//Disabling joint transmission evaluation optimization
MOD NRDUCELLCOMP: NrDuCellId=0, DCompAlgoSwitch=JT_JUDGE_OPT_SW-0;
MOD NRDUCELLCOMP: NrDuCellId=1, DCompAlgoSwitch=JT_JUDGE_OPT_SW-0;
//Disabling SRS-based weight enhancement for joint transmission
MOD NRDUCELLCOMP: NrDuCellId=0, DCompAlgoSwitch=JT_SRS_WEIGHT_ENH_SW-0;
MOD NRDUCELLCOMP: NrDuCellId=1, DCompAlgoSwitch=JT_SRS_WEIGHT_ENH_SW-0;
//Disabling rank adaptation for joint transmission
MOD NRDUCELLCOMP: NrDuCellId=0, DCompAlgoSwitch=JT_RANK_ADAPT_SW-0;
MOD NRDUCELLCOMP: NrDuCellId=1, DCompAlgoSwitch=JT_RANK_ADAPT_SW-0;
//Disabling RMSI resource reservation optimization
MOD NRDUCELLPDSCH: NrDuCellId=0, DPdschAlgoSwitch=RMSI_RES_RSV_OPT_SW-0;
MOD NRDUCELLPDSCH: NrDuCellId=1, DPdschAlgoSwitch=RMSI_RES_RSV_OPT_SW-0;
//Disabling phase compensation for Co-MM
MOD NRDUCELLCOLLABSERV: NrDuCellId=0, CommAlgoSwitch=COMM_PHASE_COMPENSATION_SW-0;
MOD NRDUCELLCOLLABSERV: NrDuCellId=1, CommAlgoSwitch=COMM_PHASE_COMPENSATION_SW-0;
//Disabling weight optimization for Co-MM
MOD NRDUCELLCOLLABSERV: NrDuCellId=0, CommAlgoSwitch=WEIGHT_OPT_SW-0;
MOD NRDUCELLCOLLABSERV: NrDuCellId=1, CommAlgoSwitch=WEIGHT_OPT_SW-0;
//Disabling coordination between joint transmission and SSB rate matching
MOD NRDUCELLCOMP: NrDuCellId=0, DCompAlgoExtSwitch=JT_SSB_RM_COORD_SW-0;
MOD NRDUCELLCOMP: NrDuCellId=1, DCompAlgoExtSwitch=JT_SSB_RM_COORD_SW-0;
//Disabling coordination between serving and cooperating cells with different numbers of PDCCH symbols
MOD NRDUCELLPDCCH: NrDuCellId=0, PdcchAlgoEnhSwitch=PDCCH_SYM_NUM_DIFF_COORD_SW-0;
MOD NRDUCELLPDCCH: NrDuCellId=1, PdcchAlgoEnhSwitch=PDCCH_SYM_NUM_DIFF_COORD_SW-0;
//Disabling optimized cooperating cell selection
MOD NRDUCELLCOMP: NrDuCellId=0, CompOptSwitch=COOP_CELL_ALLOC_OPT_SW-0;
MOD NRDUCELLCOMP: NrDuCellId=1, CompOptSwitch=COOP_CELL_ALLOC_OPT_SW-0;
//Disabling priority adjustment for coordination and UE power saving
MOD NRDUCELLCOLLABSERV: NrDuCellId=0, CommAlgoSwitch=COOR_UE_PWR_SAVE_PRI_ADJ_SW-0;
MOD NRDUCELLCOLLABSERV: NrDuCellId=1, CommAlgoSwitch=COOR_UE_PWR_SAVE_PRI_ADJ_SW-0;
//Restoring the setting of the multi-carrier calibration factor
MOD GNODEBPARAM: MulticarrierCalibFactor=1;
//Turning off the calibration robustness switches and restoring the calibration filtering coefficient
MOD GNBCLUSTER: ClusterId=0, CalibRobustSw=CALIB_FILTER_SW-0&CALIB_CELL_ROBUST_SW-0,
CalibFilterCoeff=90;
//Disabling delayed first-packet scheduling
MOD NRDUCELLCOLLABSERV: NrDuCellId=0, CommAlgoSwitch=FIRST_PACKET_SCH_DELAY_SW-0;
MOD NRDUCELLCOLLABSERV: NrDuCellId=1, CommAlgoSwitch=FIRST_PACKET_SCH_DELAY_SW-0;
//Disabling enhanced Co-MM
MOD NRDUCELLCOLLABSERV: NrDuCellId=0, MultiCellMimoSwitch=INTRA_GNB_SU_COMM_ENH_SW-0;
MOD NRDUCELLCOLLABSERV: NrDuCellId=1, MultiCellMimoSwitch=INTRA_GNB_SU_COMM_ENH_SW-0;
//Disabling Co-MM
MOD NRDUCELLCOLLABSERV: NrDuCellId=0, MultiCellMimoSwitch=INTRA_GNB_SU_COMM_SW-0;
MOD NRDUCELLCOLLABSERV: NrDuCellId=1, MultiCellMimoSwitch=INTRA_GNB_SU_COMM_SW-0;
//Removing the gNodeB MIMO cluster cell configurations
RMV GNBMIMOCLUSTERCELL: ClusterId=0, Mcc="123", Mnc="45", gNodeBId=0, CellId=0;
RMV GNBMIMOCLUSTERCELL: ClusterId=0, Mcc="123", Mnc="45", gNodeBId=0, CellId=1;
//Removing the gNodeB cluster
RMV GNBCLUSTER: ClusterId=0;
```

4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.4.2 Activation Verification

Over-the-Air Channel Calibration Result Query

Run the **DSP GNB MIMO CLUSTER CELL** command to check the over-the-air channel calibration result. If the value of **Cluster Cell Status** is **Normal**, the cells have entered over-the-air channel calibration. If the value of **Inter-TRP Calibration Result** is **Success**, the over-the-air channel calibration is successful; otherwise, the calibration is unsuccessful.

Co-MM Activation Verification

- Using counters
Observe the values of the **N.IntragNB.DL.JT.SchUser.Sum** and **N.IntragNB.DL.NCJT.SchUser.Sum** counters. If **N.IntragNB.DL.JT.SchUser.Sum** minus **N.IntragNB.DL.NCJT.SchUser.Sum** is greater than 0, Co-MM has taken effect.
- Using monitoring items
On the MAE-Access, choose **Monitor > Signaling Trace > Signaling Trace Management > NR > Cell Performance Monitoring > CoMM Monitoring**. Check the values of **CoMM Schedule Users Num** and **CoMM Schedule RB Num**. If both of them are non-zero values, Co-MM has taken effect.

4.4.3 Network Monitoring

You can evaluate the gains of Co-MM by comparing the values of certain performance indicators before and after this function is enabled.

- The downlink throughput of UEs in overlapping areas will increase.
Downlink throughput of UEs in overlapping areas (Mbit/s) = $(\text{N.ThpVol.DL.CellOverlap} - \text{N.ThpVol.DL.LastSlot.CellOverlap}) / \text{N.ThpTime.DL.RmvLastSlot.CellOverlap} \times 1000$. [Table 4-8](#) lists the related performance counters.
- The KPI **User Downlink Average Throughput (DU)** will increase.

Table 4-8 Downlink performance counters for UEs in overlapping areas

Counter ID	Counter Name
1911820651	N.ThpVol.DL.CellOverlap
1911823024	N.ThpVol.DL.LastSlot.CellOverlap
1911823026	N.ThpTime.DL.RmvLastSlot.CellOverlap

NOTICE

Throughput-related counters for UEs in overlapping areas are stepped only when the **INTRA_GNB_OVERLAPPING_MEAS_SW** option of the **NRDUCellAlgoSwitch.CompSwitch** parameter is selected. In addition, the **NRCellComp.OverlappingRsrpOffset** parameter needs to be configured, and its value must be the same as the value of either **NRCellComp.UlCompRsrpOffset** or **NRCellComp.DlCompRsrpOffset**.

When Co-MM is enabled, it is recommended that the **NRCellComp.OverlappingRsrpOffset** and **NRCellComp.DlCompRsrpOffset** parameters be set to the same value.

To compare the throughput before and after Co-MM is enabled, you are advised to perform the preceding configuration one week before Co-MM is enabled. This ensures that sufficient statistics can be collected for comparison. For detailed operations, see [4.4.1.2 Using MML Commands](#).

5 Parameters

The following hyperlinked EXCEL files of parameter reference match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and the reserved parameters that fall into disuse in the current R version.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

- Step 1** Open the EXCEL file of parameter reference.
- Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

- Step 1** Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter. View its information, including the meaning, values, and impacts.

----End

6 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

- Step 1** Open the EXCEL file of performance counter reference.
- Step 2** On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All counters related to the feature are displayed.

----End

7 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

8 Reference Documents

- 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception"
- 3GPP TS 38.331: "NR; Radio Resource Control (RRC) protocol specification"
- *Carrier Aggregation*
- *Remote Interference Management (Low-Frequency TDD)*
- *Energy Conservation and Emission Reduction*
- *Multi-RAT Coordinated Energy Saving*
- *MIMO (TDD)*
- *Coordinated Interference Management (Low-Frequency TDD)*
- *Multi-Operator Sharing*
- *Channel Management*
- *Downlink Scheduling*
- *Network Slicing*
- *Interference Avoidance*
- *LTE and NR Power Allocation*
- *Hyper Cell*
- *Cell Combination*
- *High Speed Mobility*
- *Standards Compliance*
- *Distributed Antenna Solutions (Low-Frequency TDD)*
- *DRX*
- *NR Inter-Carrier Dynamic Power Allocation*
- *MRO*
- *UE Power Saving*
- *FWA*
- *Multi-Frequency Smart Aggregation*
- *Indoor Coverage*
- *Synchronization*
- *CA Experience Improvement*

- *Positioning*
- *Extended Cell Range*
- *License Control Item Lists*
- *Uplink Scheduling*
- *Massive MIMO iBeam (Low-Frequency TDD)*
- *Fusion Cell (Low-Frequency TDD)*
- *RedCap*
- *BBU Interconnection*